

Comparison of the Clinical Frailty Score and ASA Score Versus ASA Score Alone in Predicting Postoperative Outcomes in Non-Cardiac Surgery Patients

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Abstract

Introduction: The ASA Score is the current reference to classify the global health of a patient before a surgery, providing a standardized estimate of perioperative risk. However, it does not take into account the capacities of the patient, like his activities. This is why clinicians created the Clinical Frailty Scale, only for patients above 65 years old. However, its incremental predictive value over the ASA score remains uncertain.

Methods: We performed analysis on the hospital stays, the degree of complication, and the mortality, on patients above 65 years old. We used two different models, such as mixed effect logistic regression, and mixed effect logistic ordinal regression. We computed the ROC Curve for predictions.

Results: The difference between models with and without CFS is modest across all outcomes. The AUC for the mortality without the CFS is 0.7552 whereas it is 0.7818 with. The confidence intervals show substantial overlap across models.

Conclusion: The ASA Score is a powerful tool that is sufficient alone, the CFS is significant and has a statistical value, but is not powerful enough to improve the predictions.

1 Introduction

Preoperative risk assessment plays a critical role in predicting and optimizing postoperative outcomes. The ASA score (American Society of Anesthesiologists Classification) is a widely used tool for assessing perioperative risk; however, it primarily focuses on systemic disease burden and does not account for a patient's overall functional status or frailty.

The Clinical Frailty Score (CFS), on the other hand, provides a broader assessment of physiological reserve and vulnerability, particularly in elderly and multimorbid patients.

Recent studies suggest that frailty is an independent predictor of adverse surgical outcomes, including prolonged hospital stays, postoperative complications, and mortality. This study investigates the incremental predictive value of combining the CFS with the ASA score for postoperative outcome prediction in elderly non-cardiac surgery patients. If the combined approach proves to be superior, it could enhance perioperative risk stratification and inform tailored interventions to improve patient outcomes.

The dataset SQUEEZE at our disposal regroups 25,675 patients.

2 Methods

2.1 Selected variables

The SQUEEZE dataset has 25,675 patients. After removing missing values from chosen variables, we have 25,054 patients.

We selected only ASA Score and CFS as input variables. The goal is to compare results with and without the frailty scale to see if it is an important variable.

We will analyze three output variables: the hospital stay, the mortality, and complications. As they are different, we will not use the same model.

2.2 Models

Mortality is a binary categorical output, we then will use mixed effect logistic regression.

We converted the hospital stays as a binary categorical output to analyze it. We separated the days number by two: patients who stayed less than a week, and others.

Finally, variable "complication" has multiple categories, from 1 to 5. This represents the degree of complication, and is ordinal. We could then use the ordinal logistic regression model. However, the number of patients in each category is really imbalanced (2 people for category 5). We will then not be able to perform a model with this number of data. We will regroup category 4 and 5, which appear as really severe complication. We also tried to transform the variable as binary, we separated the variables by two, one with complication 1 and 2 and the other with complication 3, 4 and 5 (small or severe complications).

2.3 Predictions / Visualization

We wanted to use the likelihood ratio test (anova) which allows us to compare the predictions quality between the model with and the model without CFS. We also wanted to analyze the AIC and BIC which are both indicators of model performance (the BIC applies a stronger penalty for each variable added than the AIC). Lower AIC and BIC values indicate better model parsimony and statistical fit. Unfortunately, we have a lot of data and a small difference between variances can be interpreted as a significant variable whereas it is not clinically significant.

So, we divided the dataset between train data (80%) and test data (20%), and performed the ROC Curve with the Delong test to have an idea of what the CFS adds to the model. For complication, we used Cohen's Kappa to have the quality of the final decision.

3 Results

3.1 Mortality

For the mortality, the AUC for the model with CFS and ASA is 0.7818 whereas it is 0.7552. The Delong test is significant ($p = 0.00065$). The CFS seems improving a bit the performance, and the confidence interval for ROC with CFS is $[0.7457-0.8179]$ whereas it is $[0.7204-0.79]$ without.

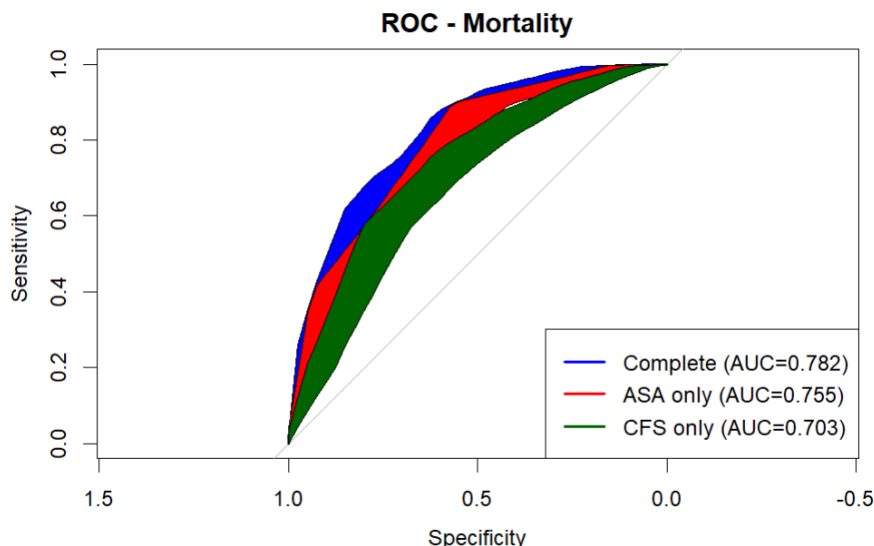


Figure 1: Comparison ROC Curve Mortality

While the overall discrimination is acceptable ($AUC = 0.782$), the incremental gain brought by the CFS over the ASA score alone remains modest (difference between $AUC = 0.027$), suggesting that the ASA score already captures the majority of the predictive signal for mortality in this population.

3.2 Hospital Stay

The hospital stay is divided in two parts, patients who are staying less than a week and patients who are staying more, short stays and long stays. The difference in AUC is minimal (0.6532 with CFS with confidence interval [0.6291-0.6772] and 0.6376, [0.6154-0.6598] without), with Delong test statistically significant ($p = 0.0025$).

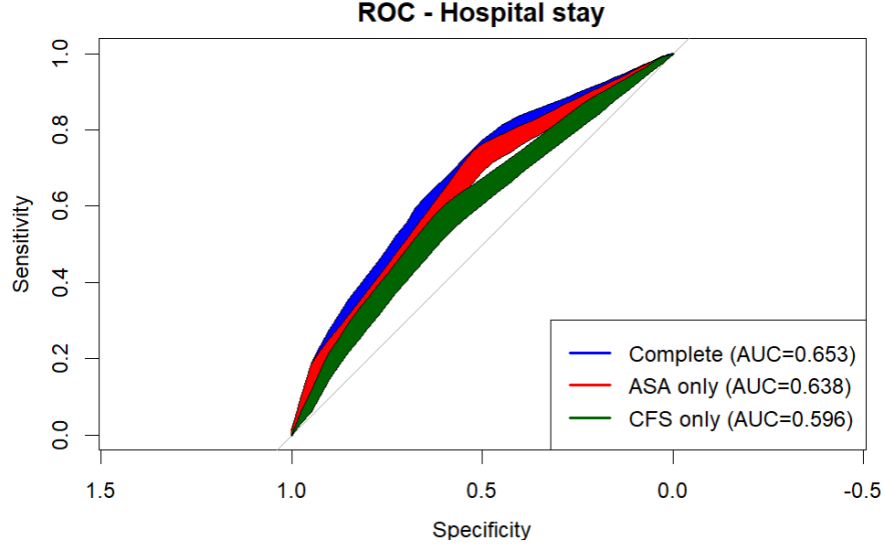


Figure 2: Comparison ROC Curve hospital stay

3.3 Complication

The variable complication has 5 categories. We first calculated Cohen's kappa, a metric that measures agreement between predicted and observed categories, corrected for chance. A value of 0 indicates that the model always predicts the same category, as a consequence of class imbalance.

Here, kappa values are near zero for all models (0.013 for the combined model, 0.019 for ASA alone, and 0 for CFS alone), indicating that none of the models achieves reliable categorical classification of complication grades. This is primarily a consequence of severe class imbalance : with approximately 60% of patients in category 1, the model systematically predicts this majority class for almost every patient, resulting in a near-zero kappa regardless of which predictors are included. The marginal difference between 0.013 and 0.019 is therefore not interpretable as a meaningful performance gap — both values reflect the same underlying phenomenon of class imbalance rather than any real difference in predictive ability between the two models. The kappa metric is particularly sensitive to this issue, which is why the AUC on the binary complication variable provides a more reliable comparison in this context.

When we performed the ROC curve on the binary variable, we obtained an AUC of 0.6304 with the CFS and 0.6296 for the model without. The p-value is not significant ($p = 0.8926$), even with the large dataset so that proves the model predicts the same with and without CFS.

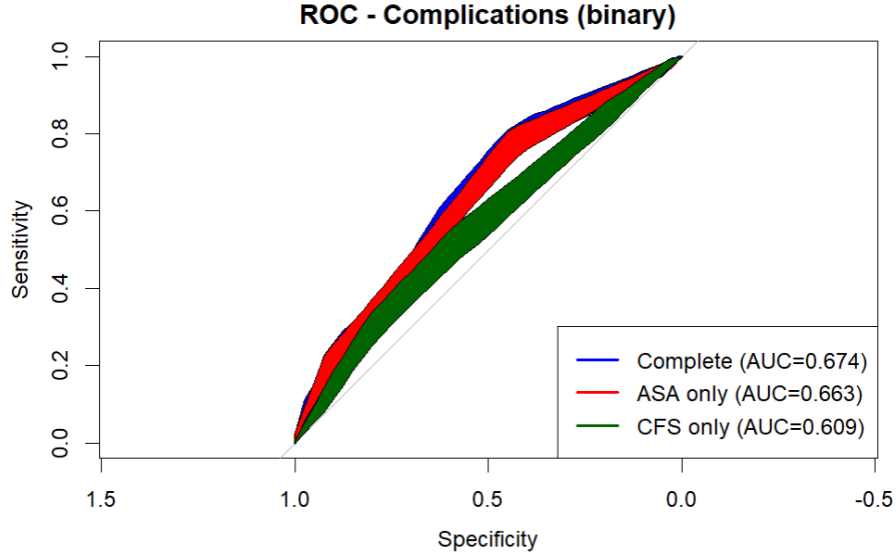


Figure 3: Comparison ROC Curve complications

Across the three outcomes studied, the addition of the CFS to the ASA score yielded statistically significant but clinically modest improvements in predictive performance, with AUC gains ranging from 0.011 (complications) to 0.027 (mortality).

4 Conclusion

To conclude, this study highlights a fundamental distinction between statistical significance and clinical relevance in predictive modeling. While the CFS reaches statistical significance in the mortality model ($p < 0.001$, DeLong), its absolute contribution — a gain of 0.027 in AUC — remains modest and may not justify its systematic addition to preoperative assessment protocols from a purely predictive standpoint.

The ASA score alone captures the vast majority of the predictive signal across all three outcomes studied (mortality, hospital stay, complications), which supports its continued use as the primary preoperative risk stratification tool.

However, several limitations temper these conclusions. The complication variable suffers from severe class imbalance, which limits the reliability of ordinal models regardless of which predictors are included.

Future studies should consider larger cohorts with more balanced complication profiles, and explore whether the CFS adds predictive value within specific surgical subgroups where frailty may be more determinant of outcomes.